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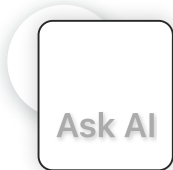
Add your data

Learn about the different ways you can overlay your data on a map.

Get started



Add map



[Add data](#)**Configure
interactions****Deploy**

This guide focuses on the process of taking geographic data that *already exists* and transforming it into a format that can be used on a web or mobile map. This could be data that you downloaded from an open data portal, data that you drew in a GIS application, aerial imagery taken by a drone, or something else.

How do I add my data to a map?

Below are four options to add your custom data to a map. If your data has to be hosted externally, skip to [Option 4](#).

Option 1: Upload data with Mapbox Studio

Best for styling your data on a map using a no-code style editor and to quickly publish map changes to your live app with minimal changes to client-side code.

This option converts your source data into a tileset that is hosted on the Mapbox platform. [Tilesets](#) are a styleable format optimized for high-performant rendering across all platforms.

Supported formats: MBTiles, KML, GPX, GeoJSON, Shapefile (zipped), CSV file, GeoTIFF

Limitations: 300 MB filesize limit.

TUTORIAL **Data in Mapbox Studio** →

Understand how to prepare and upload geospatial data in Mapbox Studio.

Option 2: Use the Mapbox Tiling Service (MTS)

Best for jobs of any size and update frequency where you want more control over how your source data is transformed and tiled. MTS is also a good option if you want to programmatically create tilesets.

Supported formats: Line-delimited GeoJSON, GeoJSON (Tilesets CLI). Note that there are several publicly available tools (like [mapshaper](#)) that convert formats like Shapefiles, KML, and CSV to GeoJSON.

Limitations: Each uploaded file must be below 20 GB. The maximum combined total size of all files that compose a tileset source is 50 GB. Read more about [MTS restrictions and limits](#).

TUTORIAL **Get started with the Mapbox Tiling Service (MTS) →**

Create and publish a custom tileset using sample data.

Option 3: Use the Uploads API

If you want to programmatically create a tileset and cannot use MTS (because GeoJSON is not a suitable format for your data or your data is in a raster format), use the Uploads API.

Supported formats: MBTiles, KML, GPX, GeoJSON, Shapefile (zipped), CSV file, GeoTIFF

Get started with the [Uploads API](#).

Option 4: Load your data at runtime

If your data lives in a third-party data-source, you can load it on the client during runtime using the `addSource` method. The examples below show how you can load an externally-hosted GeoJSON on web or mobile.

Web

iOS

Android

```
map.on('load', () => {  
  map.addSource('earthquakes', {
```

```
type: 'geojson',
// Use a URL for the value for the data property.
data: 'https://docs.mapbox.com/mapbox-gl-js/assets/earthquakes.geojson'
});

map.addLayer({
  'id': 'earthquakes-layer',
  'type': 'circle',
  'source': 'earthquakes',
  'paint': {
    'circle-radius': 4,
    'circle-stroke-width': 2,
    'circle-color': 'red',
    'circle-stroke-color': 'white'
  }
});

});
```

Dive even deeper with these code examples:

Web	iOS	Android
Load data from an external GeoJSON file Add a third party vector tile source Add an external Web Map Service layer to your map	Load data from an external GeoJSON file Add a third party vector tile source Add a third party raster tile source	Load data from an external GeoJSON file Add a third-party vector tile source Add a third-party raster tile source

How do I style my data?

You can either style your data at runtime or in Studio.

Runtime Styling: This option is ideal for event-based styling (for example, if a user hovers over data and it changes color). This option is also good if you want styling variants but don't want to support multiple styles in Studio.

Styling in Studio: This option is ideal if you want to style your data without having to alter client-side code or redeploy your application. It also keeps all your map styling in one place.

Style your data dynamically at runtime

After adding your data to a new layer, style it using one of the available [paint properties](#) for that layer type.

The example snippet below show how you can style custom line data on web with Mapbox GL JS:

```
map.addLayer({
  'id': 'route',
  'type': 'line',
  'source': 'route',
  'layout': {
    'line-join': 'round',
    'line-cap': 'round'
  },
  'paint': {
    'line-color': '#888',
    'line-width': 8
  }
});
```

Style your map in Studio

Any data uploaded to our platform is accessible from within Studio for quick styling. Changes you publish in Studio will be reflected in your mobile and web applications with no additional code changes needed.

Learn more about [styling in Studio](#).

How much will it cost?

Tilessets uploaded in Mapbox Studio are free, and do not incur processing or hosting charges. Tilessets uploaded using the Mapbox Tiling Service (MTS) or the Uploads API are billed by [tileset hosting and tileset processing](#).

Additional Developer Resources

 Mapbox Developer Discord



Developer Cheatsheet



Mapbox Support



Ask AI

Was this page helpful?

Yes

No

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